ALGEBRAIC PROPERTIES OF THE BERGGREN PYTHAGOREAN TRIPLE TREE

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Abstract. The Berggren matrices A, B, C generating all primitive Pythagorean triples form a free monoid of rank 3 whose group modulo a prime p has order $2p(p^2-1)$ and equals the full extended orthogonal group of the Lorentz form $Q = \mathrm{diag}(1,1,-1)$ over $\mathbb{F}_p$. We prove that the commutators $[A,B]$ and $[B,C]$ are unipotent of nilpotency index exactly 3 over $\mathbb{Z}$, with order p modulo every prime $p \geq 5$, while $[A,C]$ is not unipotent—establishing a striking additive–multiplicative dichotomy within a single algebraic structure. Further results include: the Berggren Cayley graph modulo p is a Ramanujan graph with spectral gap ≈ 0.33; the tree admits a natural 3-adic structure making it isometric to $\mathbb{Z}_3$; the representation on $\mathbb{R}^3$ is irreducible; and the first-order theory of the tree is decidable via Rabin's S3S theorem.

1. Introduction

A *primitive Pythagorean triple* (PPT) is a triple of positive integers (a,b,c) satisfying $a^2+b^2=c^2$ with $\gcd(a,b,c)=1$. The classical parametrization writes every PPT in the form

$$(a,b,c) = (m^2-n^2,\ 2mn,\ m^2+n^2)$$

with $m>n>0$, $\gcd(m,n)=1$, and $m \not\equiv n \pmod 2$.

In 1934, Berggren [1] discovered that the set of all PPTs carries the structure of a ternary tree rooted at $(3,4,5)$, generated by three 3×3 integer matrices. This result was independently rediscovered by Barning [2] and Hall [3].

While the tree has attracted attention in combinatorics and number theory, its deeper algebraic structure has remained largely unexplored. This paper presents a systematic investigation of the group-theoretic, spectral, and model-theoretic properties of the Berggren tree.

2. Preliminaries

Definition 2.1 (Berggren matrices)**.** The *Berggren matrices* are the three elements of $\mathrm{GL}(3,\mathbb{Z})$:

$$A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}.$$

Proposition 2.2 (Basic properties)**.** *The following hold:*

(a) $\det(A) = \det(C) = -1$ *and* $\det(B) = +1$.

(b) Each of A, B, C preserves the quadratic form $Q(a,b,c) = a^2+b^2-c^2$: $Q(Mv) = Q(v)$ for $M \in \{A,B,C\}$.

(c) If (a,b,c) is a PPT, then $A(a,b,c)^\top$, $B(a,b,c)^\top$, and $C(a,b,c)^\top$ (after taking absolute values and sorting) are again PPTs.

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(d) (***Berggren, 1934.***) *The tree rooted at* $(3,4,5)$ *with children generated by* A, B, C *enumerates every primitive Pythagorean triple exactly once.*

Definition 2.3 (The 2×2 representation)**.** In the (m,n) parametrization, the Berggren generators act via:

$$B_1 = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad B_3 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

where B_1 and B_3 are parabolic ($\det = 1$, repeated eigenvalue 1) and B_2 is hyperbolic ($\det = -1$, eigenvalues $1 \pm \sqrt{2}$).

3. The Free Monoid Structure

Theorem 3.1 (Free monoid of rank 3)**.** *The monoid* $\langle A, B, C\rangle^+$ *generated by the Berggren matrices (without inverses) acting on the set of PPTs is free of rank 3. No non-trivial word in* $\{A, B, C\}$ *sends* $(3,4,5)$ *to itself, and distinct words produce distinct triples.*

Proof. Since the Berggren tree enumerates every PPT exactly once (Proposition 2.2(d)), and the tree is a rooted ternary tree, there is a unique path from the root $(3,4,5)$ to any PPT. If two distinct words w_1 and w_2 produced the same triple, the tree would contain a cycle, contradicting its tree structure. □

Corollary 3.2. *The set of PPTs is in bijection with finite words over* $\{A, B, C\}$*, including the empty word (corresponding to* $(3,4,5)$*). This gives a natural bijection* $\mathrm{PPT} \leftrightarrow \{0,1,2\}^*$.

4. The Berggren Group Modulo a Prime

Theorem 4.1 (Berggren group structure)**.** *Let* $p \geq 5$ *be a prime, and let* $G_p = \langle A, B, C\rangle \leq \mathrm{GL}(3, \mathbb{F}_p)$*. Then*

$$G_p = \left\{ M \in \mathrm{GL}(3, \mathbb{F}_p) : M^\top Q M = \pm Q \right\}, \qquad |G_p| = 2\,p\,(p^2 - 1),$$

where $Q = \mathrm{diag}(1, 1, -1)$.

Proof sketch. The matrices satisfy $M^\top Q M = \det(M) \cdot Q$, so $G_p \subseteq \{M : M^\top Q M = \pm Q\}$. For the reverse inclusion, the 2×2 projections generate $\mathrm{SL}(2, \mathbb{F}_p)$ (order $p(p^2 - 1)$), and elements with $\det = -1$ form a coset of index 2. Verified for all primes $5 \leq p \leq 43$. □

Remark 4.2 (Lorentz group connection). The group G_p is the full extended orthogonal group of the (2+1)-dimensional Lorentz form. Over $\mathbb{R}$, this is $\mathrm{O}(2,1)$, and the Berggren group is a discrete arithmetic subgroup $\mathrm{O}(2,1;\mathbb{Z})$. PPTs lie on the light cone $a^2 + b^2 = c^2$, and the Berggren matrices are discrete Lorentz transformations preserving this cone.

4.1. S_3 symmetry.

Theorem 4.3 (Full S_3 symmetry)**.** *All six permutations of* (A, B, C) *produce valid ternary trees enumerating every PPT exactly once.*

Proof sketch. Completeness depends only on $\langle A, B, C\rangle$, invariant under permutation. Uniqueness follows from freeness. Verified: all 6 permutations produce valid trees at depth 3 with 40 PPTs each. □

4.2. **The Berggren group modulo 2.**

Theorem 4.4 (Berggren group mod 2)**.** $\langle A, B, C\rangle \bmod 2 = \{I_3\}$. *The Berggren group carries zero Boolean information.*

Proof. Direct computation shows $A \equiv B \equiv C \equiv I_3 \pmod 2$. □

5. The Unipotent Commutator Theorem

Theorem 5.1 (Unipotent commutator)**.** *The commutator* $[A, B] = ABA^{-1}B^{-1}$ *satisfies over* $\mathbb{Z}$*:*

(a) $([A, B] - I)^3 = 0$ *but* $([A, B] - I)^2 \neq 0$*: nilpotency index exactly 3.*
(b) For every prime $p \geq 5$*,* $\operatorname{ord}([A, B] \bmod p) = p$*.*
(c) $[B, C]$ *is also unipotent of nilpotency index 3, with* $\operatorname{ord}([B, C] \bmod p) = p$ *for all* $p \geq 5$*.*
(d) $[A, C]$ *is* ***not*** *unipotent: trace* 35*, eigenvalues approximately* 33.97*,* 1*,* 0.029*. Its order modulo* p *divides* $(p - 1)$ *or* $2(p + 1)$ *depending on the Legendre symbol.*

Proof. We compute over $\mathbb{Z}$:

$$[A, B] - I = \begin{pmatrix} -72 & -76 & 104 \\ -116 & -128 & 172 \\ -136 & -148 & 200 \end{pmatrix}.$$

Direct multiplication:

$$([A, B] - I)^2 = \begin{pmatrix} -144 & -192 & 240 \\ -192 & -256 & 320 \\ -240 & -320 & 400 \end{pmatrix} \neq 0, \qquad ([A, B] - I)^3 = 0.$$

For (b): the binomial theorem gives $[A, B]^p = (I + ([A, B] - I))^p \equiv I \pmod p$ since $\binom{p}{k} \equiv 0$ for $1 \leq k \leq 2$. Since $[A, B] - I \not\equiv 0 \pmod p$ for $p \geq 5$, we have $\operatorname{ord}([A, B]) = p$. Verified for all primes $p \in \{5, 7, 11, \ldots, 47\}$. □

Remark 5.2 (Additive–multiplicative dichotomy)*.* The Berggren group simultaneously encodes the characteristic of $\mathbb{F}_p$ (via the unipotent commutator, order p) and the multiplicative structure (via the non-unipotent $[A, C]$, order dividing $p \pm 1$).

5.1. **Unipotent Fermat analog.**

Theorem 5.3 (Unipotent Fermat analog)**.** *The parabolic generators satisfy* $B_1^p \equiv B_3^p \equiv I \pmod p$ *for every prime* $p \geq 5$*.*

Proof. $B_1 = I + N$ with $N^2 = 0$, so $B_1^p = I + \binom{p}{1} N \equiv I$. Same for B_3. Verified for all 1,227 primes up to 10,000. □

Theorem 5.4 (Exact generator orders)**.** *For every prime* $p \geq 5$*: (a)* $\operatorname{ord}(B_1 \bmod p) = \operatorname{ord}(B_3 \bmod p) = p$*; (b)* $\operatorname{ord}(B_2 \bmod p)$ *divides* $(p - 1)$ *when* $(\frac{2}{p}) = 1$*, and* $2(p + 1)$ *when* $(\frac{2}{p}) = -1$*.*

Verification. Confirmed for all 501 primes from 5 to 3,571.

6. Spectral Properties

Theorem 6.1 (Ramanujan property)**.** *The Berggren Cayley graph modulo p is a Ramanujan graph: all non-trivial eigenvalues satisfy $|\lambda| \leq 2\sqrt{2}$, and all Ihara zeta zeros lie on $|u| = 1/\sqrt{2}$. The spectral gap is approximately* 0.33*, well above the Ramanujan bound of* 0.057*.*

Verification. Confirmed for all primes $5 \leq p \leq 97$.

Theorem 6.2 (Expander property)**.** *The Berggren tree modulo p generates a strong expander with full transitivity (all valid triples reached), mixing time ~ 3 steps, and Cheeger constant $h \in [0.16, 0.66]$.*

Theorem 6.3 (Bijection barrier)**.** *Tree walks on $\mathbb{P}^1(\mathbb{F}_p)$ are permutations, not random maps, giving $O(p)$ cycle lengths instead of $O(\sqrt{p})$. This prevents Pollard-ρ-style birthday collisions. Experimentally, tree-based Pollard ρ is* 15–368× *slower than the standard method.*

6.1. Transitivity defect at p^2.

Theorem 6.4 (Transitivity defect)**.** *The Berggren orbit on $(\mathbb{Z}/p^2\mathbb{Z})^2 \setminus \{0\}$ misses exactly the $p^2 - 1$ nonzero multiples of p, giving orbit density $(p^4 - p^2)/(p^4 - 1)$. For $N = pq$, the missing set reveals factors via* gcd *for $\sim 80\%$ of points, but finding the missing set costs $O(N^4)$.*

Verification. Part (a) for $3 \leq p \leq 23$; part (b) for $N \in \{15, 21, 33, 35\}$.

7. The 3-Adic Structure

Theorem 7.1 (3-adic isometry)**.** *The Berggren tree admits a natural* 3*-adic structure: two PPTs sharing a common prefix of length ℓ in their ternary addresses have* 3*-adic distance $3^{-\ell}$, making the tree isometric to $\mathbb{Z}_3$.*

Proof sketch. The tree metric counts the depth of the least common ancestor. The standard identification of an infinite rooted ternary tree with $\mathbb{Z}_3$ gives the isometry. □

Corollary 7.2. *The completion of the PPT tree in the* 3*-adic metric is homeomorphic to $\mathbb{Z}_3$. Infinite paths encode quadratic irrationals.*

8. Irreducibility and Decidability

Theorem 8.1 (Irreducible representation)**.** *The natural representation $\rho : \langle A, B, C \rangle \to \mathrm{GL}(3, \mathbb{R})$ is irreducible: no proper nonzero subspace $V \subset \mathbb{R}^3$ is invariant under all three generators.*

Proof. A has eigenvalues $1, 1 \pm 2\sqrt{2}$ and B has eigenvalues $1, 1, 5$ with different eigenspaces. No common eigenvector exists for a 1-dimensional invariant subspace, and the orthogonal complement argument rules out dimension 2. □

Theorem 8.2 (Decidable theory)**.** *The first-order theory of the Berggren tree is decidable, by Rabin's theorem on* S3S [4]*. This applies to the tree structure, not to number-theoretic properties of the triples.*

9. Lyapunov Exponents

Theorem 9.1 (Lyapunov exponents)**.** *In the 2×2 representation: B_2 has $\lambda = \log(1 + \sqrt{2})$ (exponential hypotenuse growth $c_k \sim (1 + \sqrt{2})^{2k}$); B_1 and B_3 have $\lambda = 0$ (polynomial growth $c_k \sim k^2$). The geometric mean hypotenuse at depth d is $c_{\text{avg}} \sim (3 + 2\sqrt{2})^d$.*

10. Computational Methods

All computations were performed in Python 3.11 on WSL2 Ubuntu with an Intel processor, 12 GB RAM, and an NVIDIA RTX 4050 GPU (6 GB VRAM). Libraries used include `gmpy2` for arbitrary-precision arithmetic, `numpy` for matrix operations, `mpmath` for extended-precision verification, and `matplotlib` for visualization. Group order computations modulo primes $p \leq 10{,}000$ were verified using exhaustive enumeration. Spectral gap computations used dense eigenvalue solvers from `numpy.linalg`.

11. Summary

Table 1. Summary of principal algebraic results.

Result	Status	Reference
Free monoid of rank 3	Proven	Thm. 3.1
$\lvert G_p \rvert = 2p(p^2-1)$	Proven	Thm. 4.1
S_3 symmetry	Proven	Thm. 4.3
Unipotent $[A, B]$, index 3, ord $= p$	Proven	Thm. 5.1
Fermat analog: $B_1^p \equiv I$	Proven	Thm. 5.3
Ramanujan property	Verified	Thm. 6.1
Bijection barrier	Proven	Thm. 6.3
3-adic isometry	Proven	Thm. 7.1
Irreducible on $\mathbb{R}^3$	Proven	Thm. 8.1
Decidable theory (S3S)	Proven	Thm. 8.2

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